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Validity of Global Positioning System Technology to Measure Maximum Velocity Sprinting in Elite Sprinters

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Abstract

Thome, M, Thorpe, RT, Jordan, MJ, and Nimphius, S. Validity of global positioning system (GPS) technology to measure maximum velocity sprinting in elite sprinters. *J Strength Cond Res* 37(12): 2438–2442, 2023—The objective of this study was to assess the concurrent validity of 10-Hz wearable Global Positioning System (GPS) technology to measure maximum velocity sprinting (V_{max}) relative to Doppler radar in elite sprinters. Data were collected from a single training session performed by elite 100 and 200 m sprinters (males: $n = 5$; 100 m best times: 10.02 ± 0.07 seconds, range: 9.94–10.10 seconds; 200 m best times: 20.29 ± 0.42 seconds, range: 19.85–20.80 seconds; females: $n = 2$; age: 28.0 ± 4.2 years; body mass: 65.8 ± 4.6 kg; 100 m best times: 11.18 ± 0.34 seconds; 200 m best times: 22.53 ± 0.04 seconds). Velocity and time data from 16 maximal, 60-m sprint efforts were recorded simultaneously with 10 Hz GPS and 47 Hz radar. Validity was assessed using Bland-Altman 95% limits of agreement (LOA) and intraclass correlation coefficient (ICC), each with respective 95% confidence intervals (CI). V_{max} measured with 10 Hz GPS demonstrated a LOA of -0.11 m·s^{-1} (-0.17, -0.05) and an ICC of 0.99 (0.98, 1.0) relative to the radar device.10 Hz GPS overestimated V_{max} by 0.11 m·s^{-1} relative to the radar but could still be considered a suitable tool for monitoring external load in elite sprinters. However, the much smaller average annual improvement in this population (~0.1–0.2%) in comparison with the ~1% overestimation reduces the utility of 10 Hz GPS to detect meaningful performance changes in maximum velocity.

Key Words: track and field, training load, athlete monitoring

Introduction

Maximum horizontal velocity (V_{max}) is a key predictor of sprint performance (12,19,20). Furthermore, the ability to accurately measure and quantify sprinting at high velocities is important as it can be used to operationalize the construct of external training load for monitoring elite sprinters. Monitoring maximal horizontal velocities is important because peak forces can exceed 4–5× bodyweight during maximal sprinting velocity (8,28). Advancing the knowledge on monitoring training loads, in this case operationalized as distances run across velocity ranges up to maximal, in elite sprinters is essential to optimize performance and avoid training prescription errors.

Ensuring that measurement devices have sufficient reliability and validity to detect small changes in V_{max} over time is a crucial factor for effective athlete monitoring (10,13). Radar timing systems that measure sprinting velocity according to the Doppler principles are considered reliable (coefficient of variation [CV]: 1.2–3.5%) and accurate ($r = 0.99$) to track high velocity sprinting (7,10,26). However, the use of a radar system to monitor sprinting velocity in an elite training environment is limited due to the location of the tripod blocking a lane on the track, the magnitude of data, and data presentation challenges. Consequently, timing gate systems are frequently used to monitor sprinting velocity (10). Despite the fact timing gates display excellent validity (bias of 0.1 m·s^{-1}) and reliability (standard error of measurement [*SEM*]: 0.02–0.03 seconds; CV: 1–2%), only average velocity estimates calculated using split times are possible, requiring the user to construct modeled velocity-time curves (23,30). Furthermore, timing gates and radar cannot track distances run across different velocities throughout an entire training session, which is a key component of monitoring external training loads.

Wearable Global Positioning System (GPS) units are widely used in team sports for assessing running velocity and running distance, as measures of external training load, and can be integrated into most training environments (24). Running distances recorded using 10 Hz GPS have demonstrated very small bias relative to radar during straight line running and team sport-specific movements (typical error as CV: 0.7–1.6%; Bias: 1.26–2.7%) (4). 10 Hz GPS velocity data have also been reported to be valid and reliable during straight line running at various velocity ranges, with improvements in accuracy as velocity increases: 1–3 m·s^{-1} (% bias: 0.6 ± 0.4; % CV: 5.3 ± 0.2), 3–5 m·s^{-1} (% bias: -0.2 ± 0.2; % CV: 3.5 ± 0.2), and 5–8 m·s^{-1} (% bias: -0.2 ± 0.2; % CV: 2.0 ± 0.12) (27). In addition to monitoring external training load, 10 and 16 Hz GPS may be viable tools for monitoring maximum velocity sprinting performance in team sport athletes, with 10 Hz units demonstrating

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a typical error of the estimate (TEE) of 1.87% (1.66–2.18%) and a mean bias of −0.78% (−1.13 to −0.42%) when compared with radar and 16 Hz GPS demonstrating a typical error of measurement (TEM) of 0.1 ± 0.01 $m \cdot s^{-1}$, a mean bias of −1.6 ± 1.1%, and a CV of 0.5 ± 0.1% (17,22). Although this level of validity may be acceptable for assessing sprint performance in team sport athletes, very small changes in performance are meaningful in elite sprinters. For example, the average annual improvement in performance is in the range of ~0.1–0.2% for this population (11). In addition, elite sprinters regularly achieve velocities higher than the values reported by Lacome et al. (17) (8.22 ± 0.23 $m \cdot s^{-1}$) and Roe et al. (22) (8.82 ± 0.55 $m \cdot s^{-1}$) in team sport athletes. Higher velocities result in additional displacement travelled between data samples, and therefore, the accuracy of GPS systems for measuring the sprinting velocities >9 $m \cdot s^{-1}$ in elite sprinters is unknown.

The purpose of this study was to assess the validity of 10 Hz wearable GPS relative to a 47 Hz radar system (criterion measure) to measure maximum velocity sprinting in elite 100 and 200 m sprinters. We hypothesized that 10 Hz GPS would demonstrate acceptable validity (ICC >0.75) at the high sprinting velocities achieved by elite sprinters for the purpose of monitoring. However, we hypothesized 10 Hz GPS would not be accurate enough to detect meaningful changes in maximum velocity as a measure of performance in this population.

Methods

Experimental Approach to the Problem

Maximal sprint efforts (n = 16, distance = 60 m; efforts per athlete = 2–3) with simultaneous recording of GPS (Catapult OptimEye S5, 10 Hz GPS, 100 Hz 3D accelerometer, 100 Hz 3D gyroscope; 100 Hz 3D magnetometer, Melbourne, VIC, Australia) and radar (Stalker ATS II, 47 Hz; Radar Sales, Minneapolis, MN) were obtained during a sprint training session. Footwear (spikes) and sprinting surface (synthetic rubber outdoor running track) were controlled between sprint efforts. The temperature on this date was 21° C, with a 0.5 $m \cdot s^{-1}$ tailwind.

Subjects

Elite sprinters, specialized in the 100 and 200 m sprint events, performed a collective total of 16 sprint trials (males: n = 5; age: 26.2 ± 2.5 years; body mass: 82.68 ± 6.94 kg; 100 m best times: 10.02 ± 0.07 seconds, range: 9.94–10.10 seconds; 200 m best times: 20.29 ± 0.42 seconds, range: 19.85–20.80 seconds; females: n = 2; age: 28 ± 4.2 years; body mass: 65.75 ± 4.6 kg; 100 m best times: 11.18 ± 0.34 seconds; 200 m best times: 22.53 ± 0.04 seconds). Athletes provided a data release consent form as part of their participation in training that provided permission to release deidentified data for research purposes. An ethics declaration for use of deidentified monitoring data was obtained through Edith Cowan University's Human Research Ethics Committee (REMS number: 2020-01346-THOME).

Procedures

The radar system was mounted on a tripod positioned 7 m behind the start line and 1 m above the ground and was aimed at the athlete's lower back (26). Velocities were calculated in the radar software package (Stalker ATS 5.0, Minneapolis, MN), and raw data files (.rda) from each sprint were exported and used for analysis.

Global Positioning System devices were activated before data collection, and the acquisition of satellite signals was confirmed. The number of satellites acquired during this session was 12, and the mean horizontal dilution of precision (HDoP) was 0.70 ± 0.03, indicating acceptable signal quality (18). A GPS vest (Catapult Sports, Melbourne, VIC, Australia) was tightly fit to each athlete with the receiver located in a pocket on the upper back, between the scapulae. Athletes were provided with their own, individual unit and wore this same GPS device for the entire session to mitigate potential error associated with interunit reliability (14).

Statistical Analyses

The raw instantaneous velocity-time data measured by radar and GPS during each sprint were fit with a generalized additive model (GAM) in R (version 4.0.2, http://www.R-project.org/; mgcv package; k = 15, method = "REML"). The gam.check() function was used to confirm model conversion and sufficient basis functions. The maximum velocity obtained from the fitted model for each device was subsequently used for analysis. The start time for each trial was determined by the first data point above zero on the velocity trace (4).

Mean and standard deviation (SD) of V_{max} values from the GPS and radar were calculated. A Bland-Altman with 95% limits of agreement (LOA) and intraclass correlation coefficients (ICCs) were selected to assess measurement error of the GPS relative to the radar measurement device (1,2). Bland-Altman plots were constructed in R statistical software (version 4.0.2, http://www.R-project.org/; BlandAltmanLeh and blandr packages). Intraclass correlation coefficients were also calculated in R (psych package) using the single fixed raters model (ICC3) (15,25). Acceptable validity for monitoring sprinting velocity was established using previously reported recommendations with ICC values interpreted as <0.5 (*poor*), 0.5–0.75 (*moderate*), 0.75–0.90 (*good*), and >0.90 (*excellent*) (15). Acceptable validity for assessing performance was interpreted relative to intended use (6). Therefore, acceptable validity was assessed relative to the average annual change in performance in this population (11).

Results

Concurrent validity data and mean (SD) V_{max} for the GPS and radar raw and GAM-fitted values are displayed in Table 1. The validity assessment using the raw radar data showed a systematic bias of 1.40 $m \cdot s^{-1}$ (0.95, 1.85) and a *moderate* ICC value of 0.73 (0.45, 0.87). However, the GAM-fitted V_{max} values showed that GPS overestimated the radar values with a smaller systematic bias of −0.11 $m \cdot s^{-1}$ (−0.17, −0.05) and an *excellent* ICC value of 0.99 (0.98, 1.0).

Discussion

We determined the error of the 10 Hz GPS device for monitoring maximum velocity as a measure to operationalize external load in elite 100 and 200 m sprinters relative to a 47 Hz radar system is acceptable. The sprinters achieved maximum velocities ranging from 8.64 to 11.05 $m \cdot s^{-1}$ (mean ± SD:

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Table 1
Mean (*SD*) of observed velocities.*†

	Observed values			
	10 Hz GPS	47 Hz radar	95% LOA	ICC
V_{max} (m·s^{-1}) (95% CI)	10.41 (0.88)	11.81 (1.35)	1.40 (0.95 to 1.85)	0.73 (0.45 to 0.87)
V_{max} (m·s^{-1}) (95% CI) using GAM smoothing	10.33 (0.88)	10.22 (0.86)	−0.11 (−0.17 to −0.05)	0.99 (0.98 to 1.0)

*LOA = limits of agreement; ICC = intraclass correlation coefficient; V_{max} = maximum velocity; CI = confidence interval.
†95% limits of agreement and intraclass correlation coefficient, with respective 95% CI, for maximum velocity sprinting comparisons between a 10 Hz Global Positioning System and a 47 Hz radar device.

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10.22 ± 0.86 m·s^{-1}), which exceeds commonly reported velocities in team sport athletes (8.22 ± 0.23 m·s^{-1}; 8.82 ± 0.55 m·s^{-1}) (17,22). Although there seemed to be a tendency toward increasing bias as velocity increased (*c.f.* Figure 1A), this data set did not contain a wide enough range of values to accurately assess if heteroscedasticity was present. However, the much smaller average annual improvement in this population (~0.1–0.2%) (11) in comparison with the ~1% overestimation reduces the utility of 10 Hz GPS to detect meaningful performance changes in maximum velocity.

Raw velocity data from the radar device returned several unrealistic values (13.5–14.3 m·s^{-1}), justifying the need for signal processing. Previous methodologies for radar or laser data processing have included the use of a 0.5 seconds moving average (16,17), down sampling to 10 Hz (27), a 5th-order polynomial function (5), a Butterworth filter with 3 Hz cutoff frequency (9), a modeled velocity-time curve using an exponential function (21,23), and the use of stock software provided by the Stalker radar system (3,4,22). When considering data filtering techniques, care must be taken to avoid aliasing the original signal by over or under smoothing. A GAM-fitted curve was selected to smooth raw data in this study as GAMs offer the flexibility to fit nonlinear relationships through spline functions (29). Splines can fit a wide range of shapes, allowing the GAM curve to follow along the natural acceleration pattern of the athlete, and fit can be confirmed using the gam.check() function (Figure 2).

Consistent with previously reported data from similar studies, V_{max} measured using 10 Hz GPS displayed acceptable validity. Using the same 10 Hz GPS device as in this study, Roe et al. (22) found a *small* TEE of 0.16 m·s^{-1} in V_{max} relative to the Stalker ATS II radar in professional rugby union players. Similarly, Kyprianou et al. (16) demonstrated a mean difference of 0.04 m·s^{-1} (90% CI: −0.03 to 0.11 m·s^{-1}) in V_{max} when comparing 10 Hz GPS to the Laveg LDM 300C 100 Hz laser in elite youth soccer players, and Beato, Devereux, et al. (4) showed a bias of 1.80 ± 1.93% between 10 Hz GPS and the Stalker ATS radar. Previous research has also demonstrated improvements in the validity and reliability of 10 Hz GPS as velocity increases (27). It is possible that the minimal changes in velocity near V_{max} contribute to the lower error in GPS measurements compared with sprint efforts involving lower velocities and higher accelerations. However, the lower error at the higher velocities of more consistent acceleration still retains an error much higher than the expected average annual improvement in sprint performance. As a result, 10 Hz GPS has the utility for monitoring sprinting velocity operationalized to monitor external load but not for assessing performance measured as maximum velocity in elite sprinters.

Previous research has shown that increasing GPS device sampling frequency improved validity and reliability up to 10 Hz, with no added benefit when increasing to 15 Hz (24). However, more recent data from Beato, Coratella, et al. (3) demonstrated bias of 2.36 ± 1.37% and 2.02 ± 1.24% in 10 and 18 Hz GPS, respectively, when compared with the Stalker ATS radar for measuring maximum velocity sprinting. Lacome et al. (17) also showed that 16 Hz GPS demonstrated a *small* TEM of 0.09 ± 0.01 m·s^{-1} in V_{max} as well as excellent reliability (CV: 0.5 ± 0.1%) when assessing the agreement between and the Stalker ATS II radar device in elite rugby sevens players (17). These studies elucidate the possibility of further improvement in the accuracy in V_{max} measures with the increased sampling frequency in newer GPS devices. Future validation work will be required as improvements are made in GPS technology to determine if higher sampling frequencies will lead to improved measurement validity and reliability.

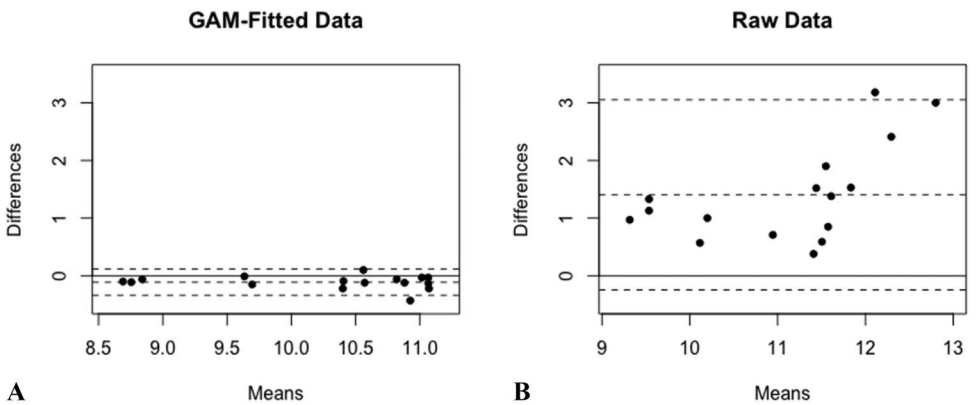

Figure 1. A and B, Bland-Altman plots displaying the mean values of the differences between GPS and Radar V_{max} values (m·s^{-1}) for GAM-fitted (A) and raw data (B).

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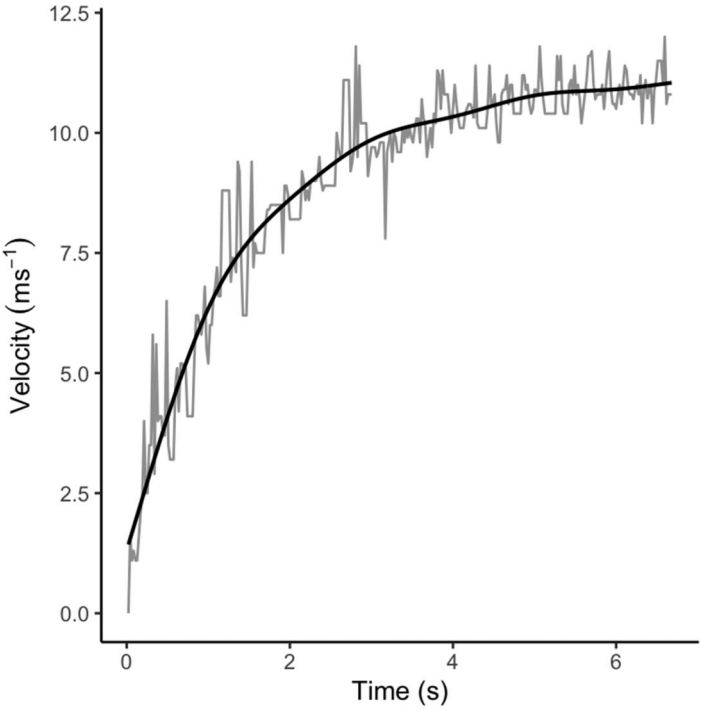

Figure 2. Example of GAM-fitted curve superimposed with raw radar signal.

Practical Applications

Coaches and practitioners working with elite sprinters capable of attaining higher than typically observed maximum sprinting velocities can be confident using 10 Hz GPS as a tool for monitoring running distances and velocities (i.e., external load). The use of 10 Hz GPS for monitoring sprinters does have an average overestimation of 0.11 m·s^{-1} relative to the criterion measure of this study. For practitioners, 10 Hz GPS is suitable for monitoring sprinters, but when evaluating performance, which is only expected to change by ~0.1–0.2% per annum in this population (11), 10 Hz GPS does not have the ability to detect meaningful changes in maximum velocity. Therefore, the ~1% overestimation in Vmax by 10 Hz GPS lacks the precision required to track performance changes in elite sprinters.

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